

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

SSC PART-I (CLASS 9) PRACTICE EXAMINATION

MATHEMATICS PAPER 20

Syllabus: Unit 6 Trigonometry and Bearing, sections 6.1 (radian measure) and 6.2 (sector of a circle), her book p117 to 121 (27 marks) + Revision: Unit 11 Basic Statistics, probability (11 marks)

Total Marks: 38

Time Allowed: 1 hour 30 minutes

Version A

HIDAYAAT (Instructions)

1. Teenon sections hal karein. Koi optional sawal nahin hai.
2. Section A mein sirf apne option ka letter likhein, jaise 1. (b). Aakhir mein gin lein ke dason ho gaye hain.
3. Har jawab ke saath
uska unit likhein
(cm, cm², radians). Unit na likhne par aadha mark katega.
4. Calculator use kar sakti hain. Jahan $\pi = 3.1416$ kaha gaya ho wahi value istemal karein.
5. **θ hamesha radians mein daalein**
, degree mein nahin. Agar zaviya degree mein diya ho to pehle badlein.
6. Probability ke jawab fraction mein likhein, aur jahan kaha jaye wahan sada tareen shakal mein.

SECTION A: MULTIPLE CHOICE QUESTIONS (10 × 1 = 10 MARKS)

#	Question	A	B	C	D
1	π radians is equal to:	360°	270°	180°	90°
2	The length of a circular arc is given by:	$l = r\theta$	$l = r/\theta$	$l = \frac{1}{2}r^2\theta$	$l = 2\pi r$
3	One degree in radian measure is equal to:	$180/\pi$	$\pi/180$	$\pi/90$	$\pi/360$
4	In the formulas $l = r\theta$ and $A = \frac{1}{2}r^2\theta$, the angle θ must be measured in:	degrees	minutes	seconds	radians
5	45° converted into radians is:	$\pi/6$	$\pi/3$	$\pi/4$	$\pi/2$
6	An arc whose length is equal to the radius subtends at the centre an angle of:	1 radian	1°	90°	π radians
7	$60^\circ 30'$ written in decimal degrees is:	60.05°	60.5°	60.3°	65°
8	[Revision] A fair coin is tossed once. The probability of getting a head is:	0	$1/4$	$1/2$	1
9	[Revision] A fair die is rolled once. The probability of getting an even number is:	$1/6$	$1/3$	$2/3$	$1/2$
10	[Revision] If $P(E) = 0.3$, then $P(\bar{E})$ is:	0.7	0.3	1.3	0

SECTION B: SHORT ANSWER QUESTIONS (3 × 4 = 12 MARKS)

Teenon sawal hal karein. Ye teenon aap ke Unit 6 Part 2 notes ke moazoo se hain.

2. (i) Convert **150°** into radians, writing your answer in terms of π . **(2)**
(ii) Convert **$7\pi/40$** radians into $D^\circ M'S''$. **(2)** **4**
3. A circle has radius **10 cm** and a sector whose central angle is **$2\pi/5$ radians**. **4**
(i) Find the length of the arc. **(2)** (ii) Find the area of the sector. **(2)**
Give each answer in terms of π and also as a decimal, with its unit.
4. (i) Define **one radian** in the words of your book. **(1)** **4**
(ii) Write the formula for the **length of an arc** and for the **area of a sector**, say what each letter stands for, and state the unit θ must be in. **(3)**

SECTION C: LONG ANSWER QUESTIONS (2 × 8 = 16 MARKS)

Dono sawal hal karein. Q.5 Unit 6 se hai aur Q.6 revision (Unit 11, probability) se.

5. Radian measure, arc length and sector area. **8**
(i) Convert **270°** into radians. Use $\pi = 3.1416$. **(2)**
(ii) Convert **$75^\circ 30' 45''$** into radians, correct to **4 decimal places**. Use $\pi = 3.1416$. **(2)**
(iii) A circle has radius **14 cm**. One of its sectors has an arc of length **22 cm**. Find the central angle of that sector in radians, and then find the area of the sector. Show the formula, the substitution and the answer with its unit. **(4)**
6. [Revision, Unit 11: Probability] **8**
(i) Define **probability** in the words of your book, and state the smallest and largest value a probability can take. **(2)**
(ii) A bag holds **5 red, 4 green** and **3 blue** balls. One ball is drawn at random. Find the probability that it is (a) red, (b) not red, (c) blue. **(3)**
(iii) A fair die is rolled once. Find the probability of getting (a) a number greater than 4, (b) a prime number. **(2)**
(iv) If $P(E) = 2/5$, find $P(\bar{E})$. **(1)**

END OF QUESTION PAPER

Unit 6, sections 6.1 and 6.2 (MCQs 1 to 7, Questions 2, 3, 4 and 5) | Revision: Unit 11 probability (MCQs 8, 9, 10 and Question 6).

Student's Name: _____ Date: _____ Marks Obtained: _____ / 38